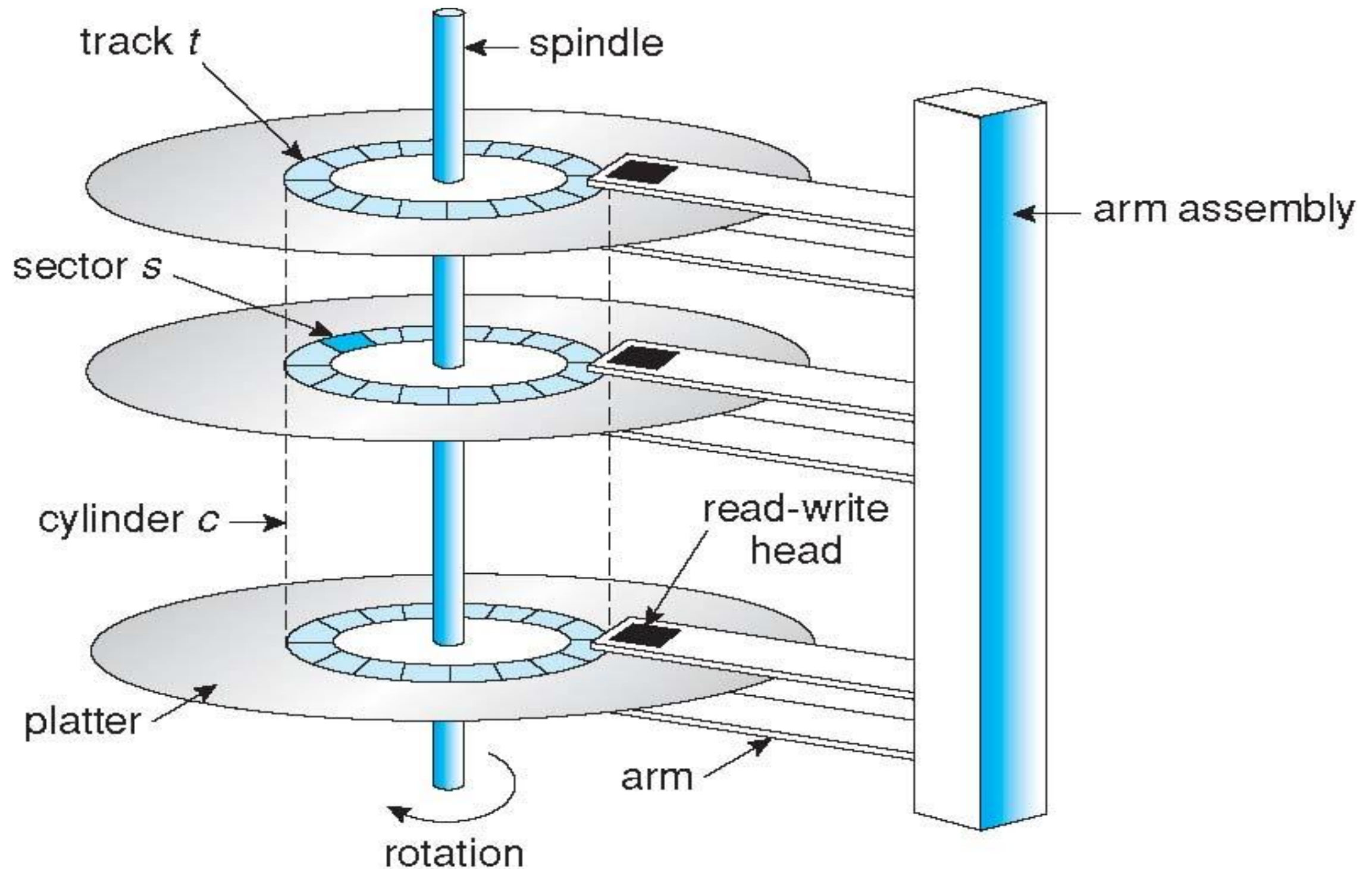


Secondary Storage Structure

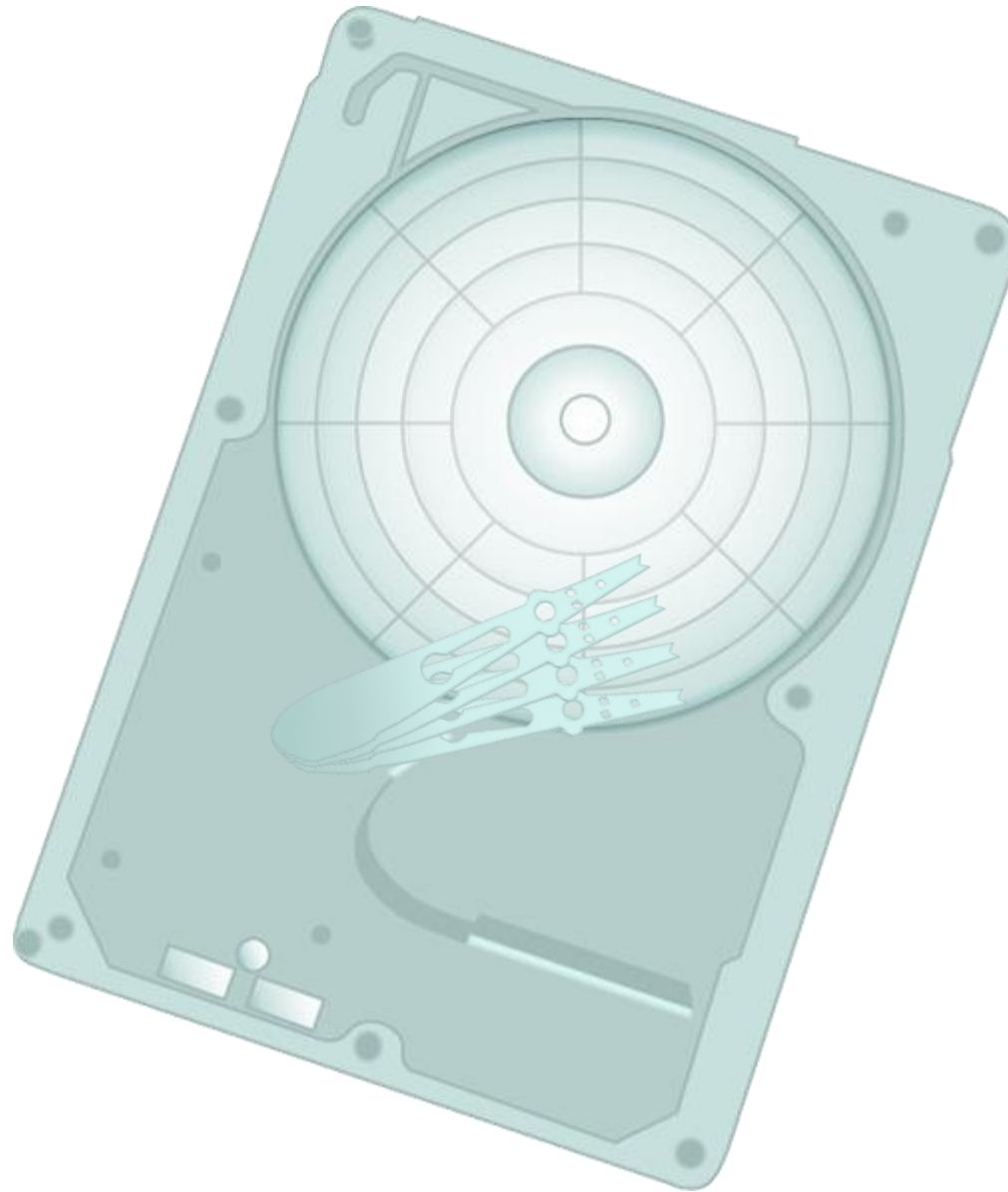
Disk Structure

- Disk drives are addressed as large 1-dimensional arrays of logical blocks, where the logical block is the smallest unit of transfer.
- The 1-dimensional array of logical blocks is mapped into the sectors of the disk sequentially.
 - * Sector 0 is the first sector of the first track on the outermost cylinder.
 - * Mapping proceeds in order through that track, then the rest of the tracks in that cylinder, and then through the rest of the cylinders from outermost to innermost.

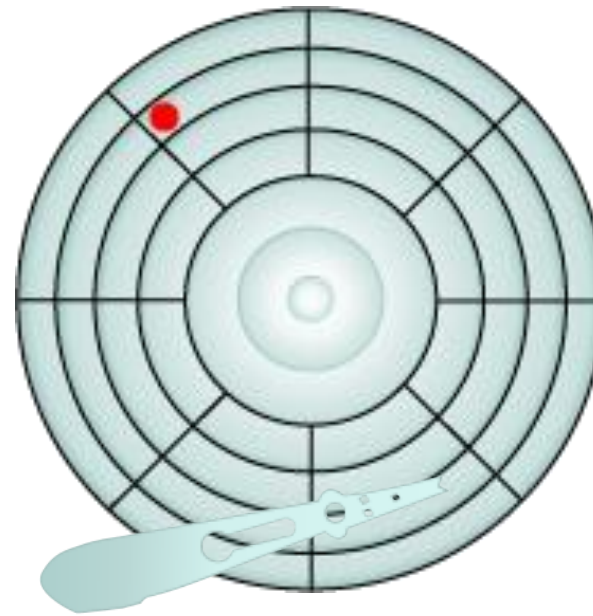
Disk Structure...



Disk Drive Performance: Seek Time



Disk Drive Performance: Rotational Speed/Latency

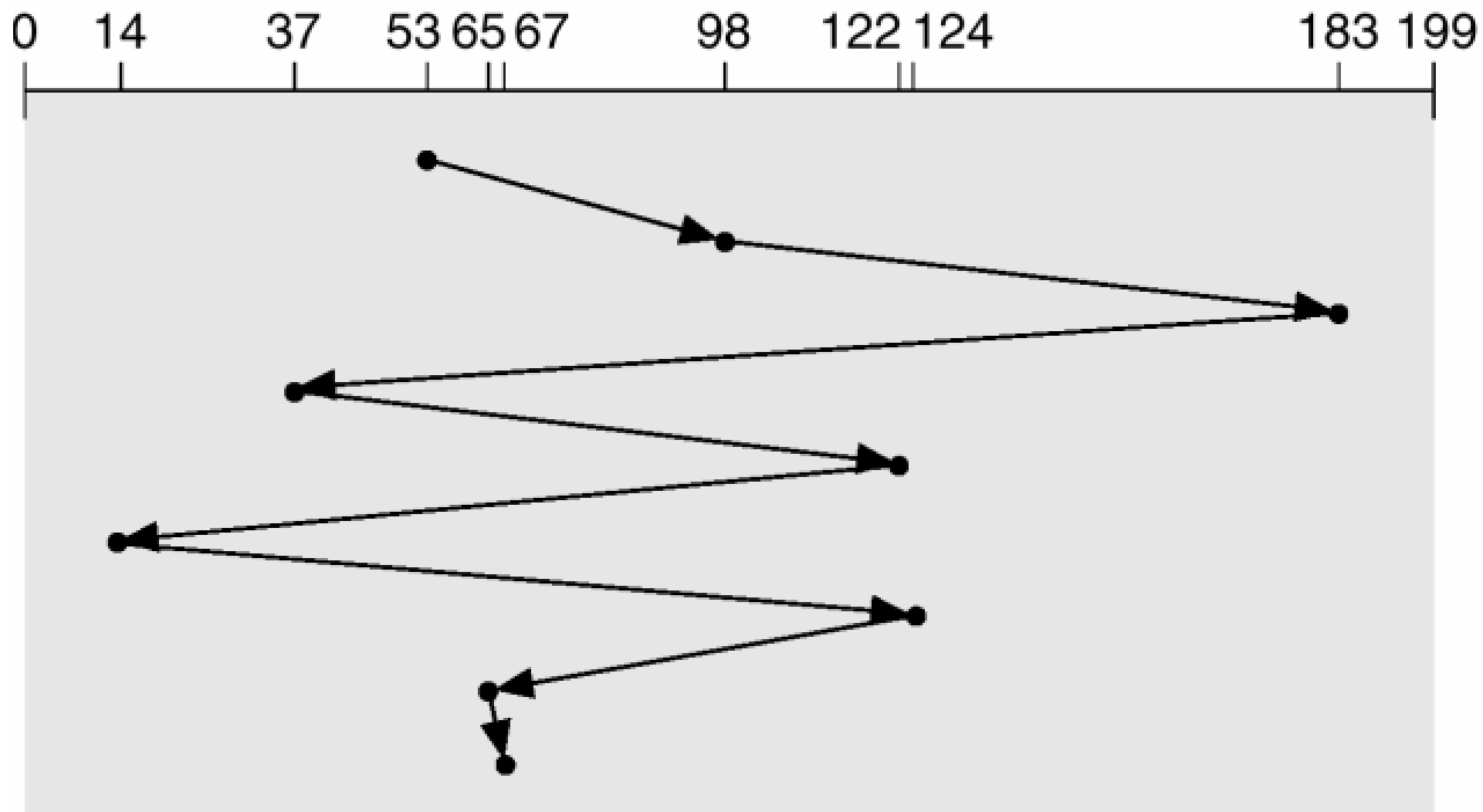


Disk Scheduling

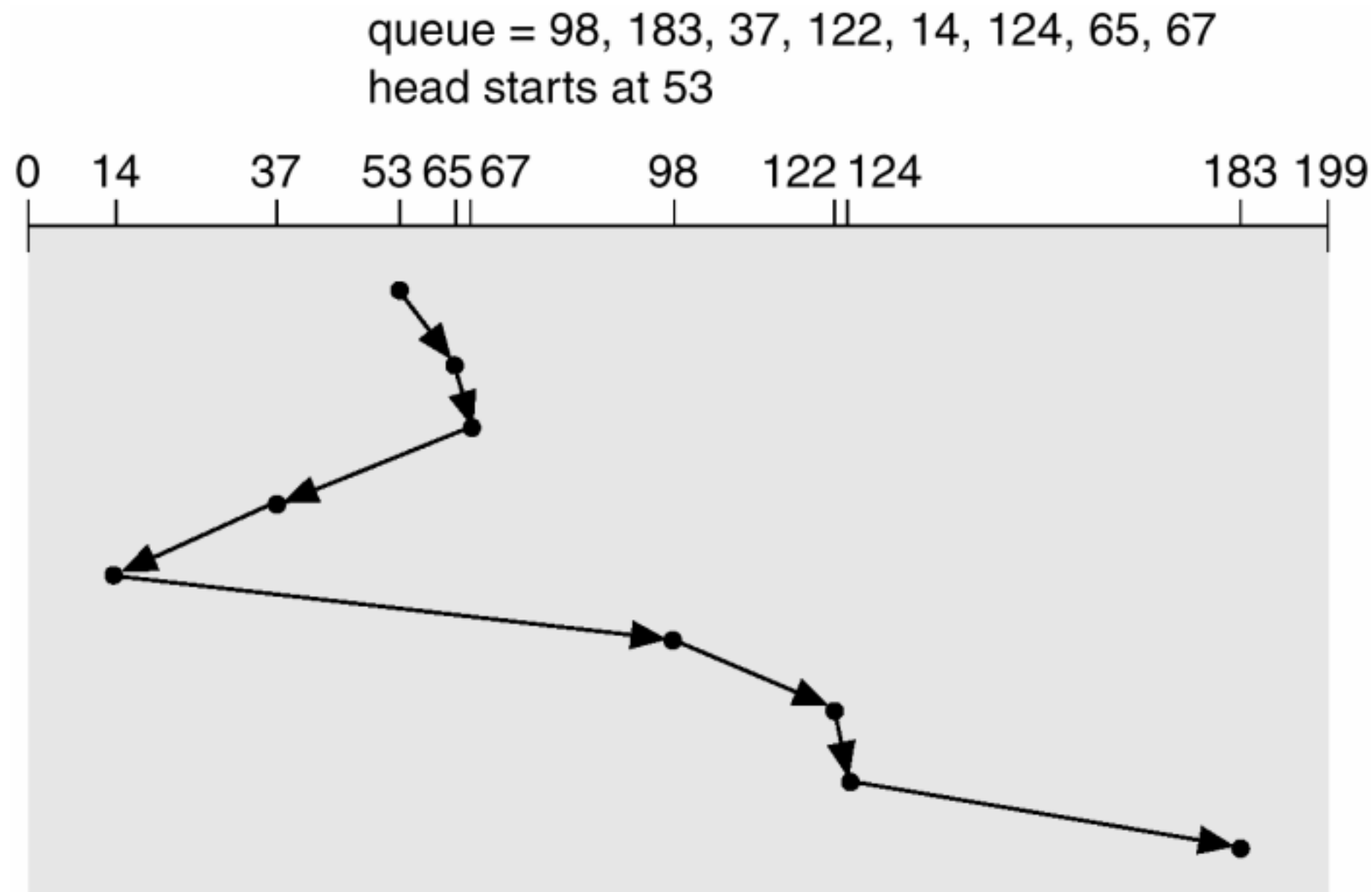
- The operating system is responsible for using hardware efficiently —for the disk drives, this means having a fast access time and disk bandwidth.
- Access time has two major components $Acc = SEEK TIME + ROTATIONAL LATENCY$
- Seek time is the time for the disk are to move the heads to the cylinder containing the desired sector. - DESIRED CYLINDER PAR JAANE M SEEK TIME LAGTA H
- Rotational latency is the additional time waiting for the disk to rotate the desired sector to the disk head. - DESIRED SECTOR PAR JAANE M ROTATIONAL LATENCY
- Minimize seek time
- Seek time » seek distance
- Disk bandwidth is the total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer.

FCFS

queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53



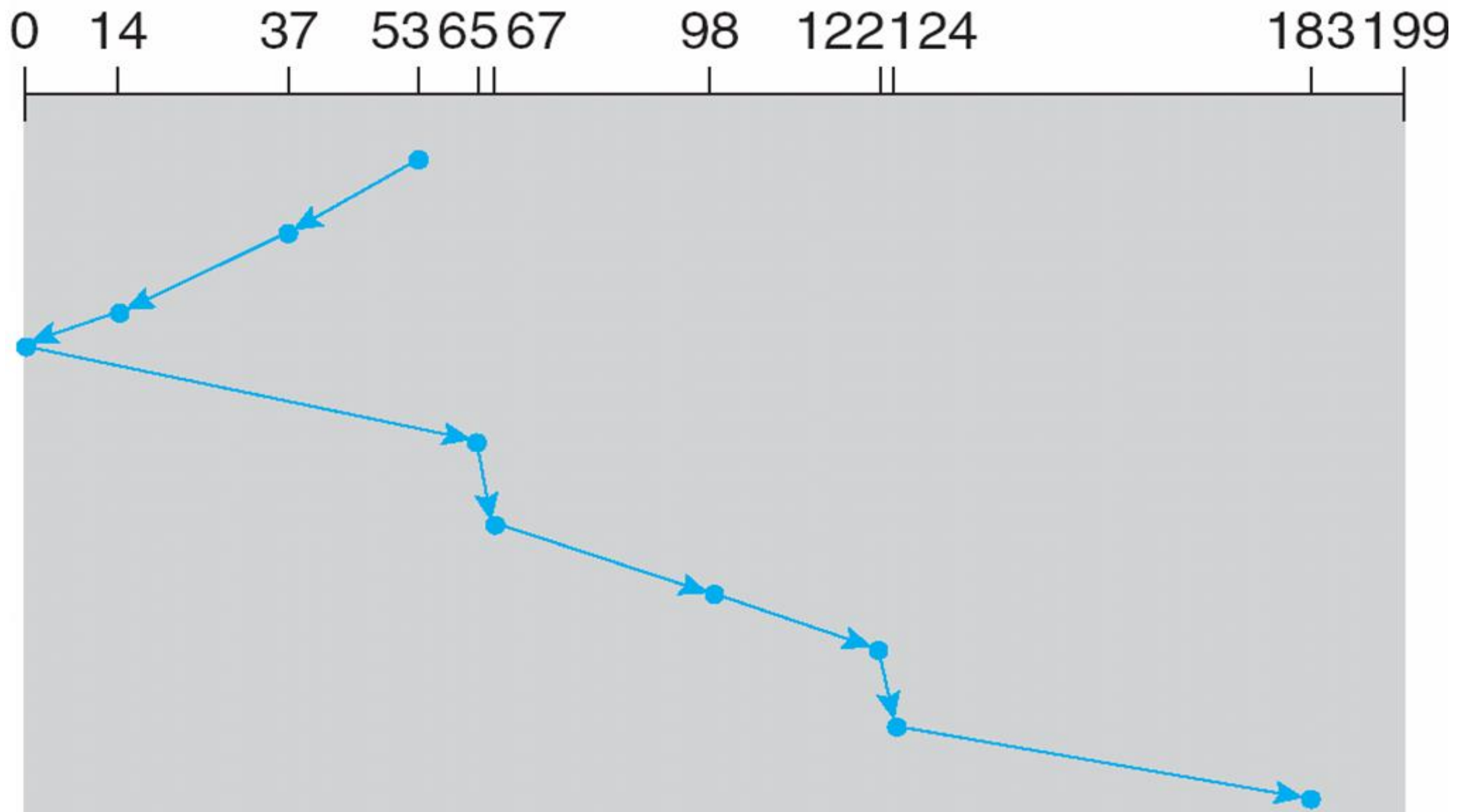
SSTF (Shortest Seek Time First)



SCAN

queue = 98, 183, 37, 122, 14, 124, 65, 67

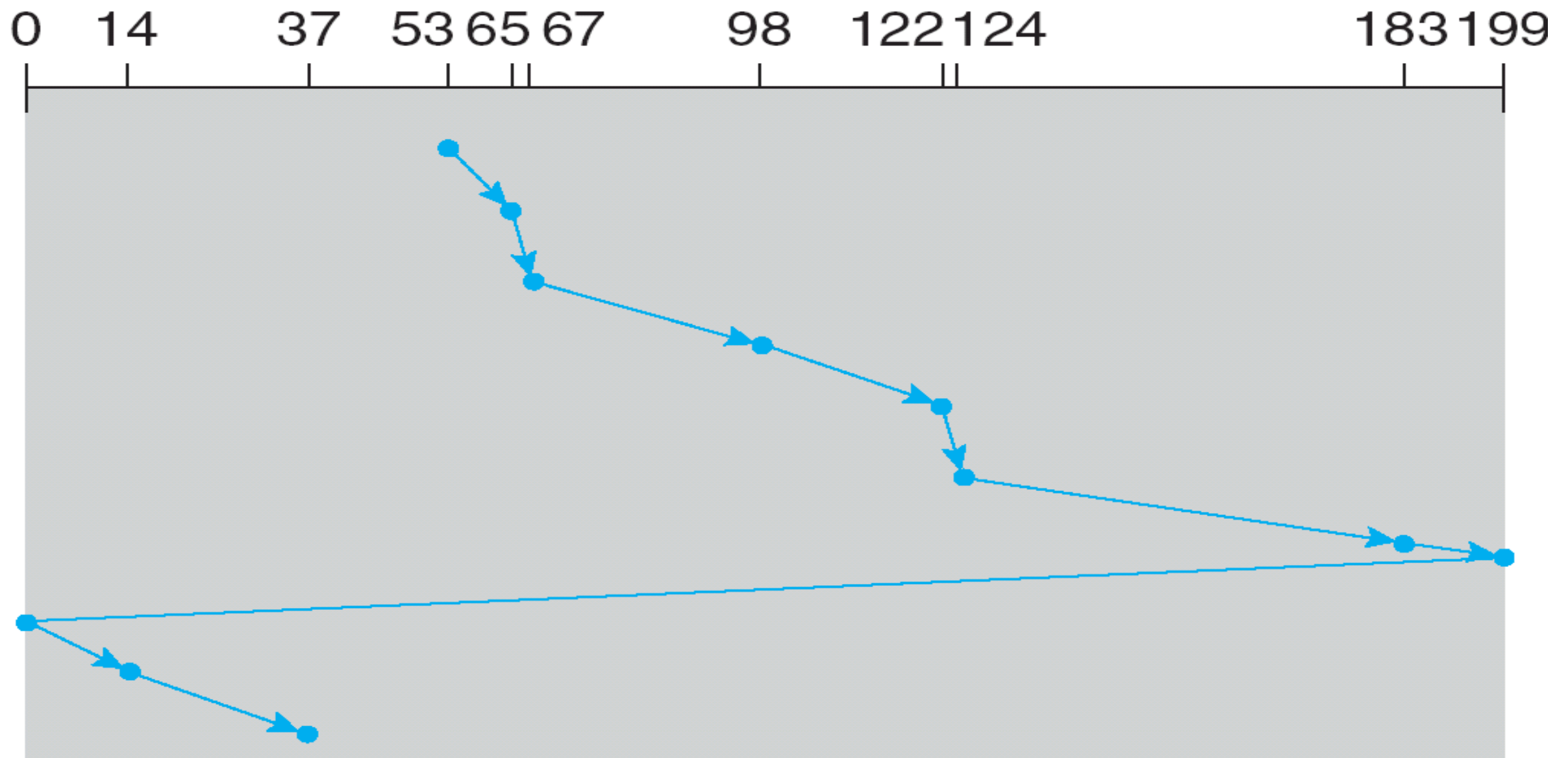
head starts at 53



C-SCAN

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53



C-LOOK

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53

